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Repelling Polysulfides Using White Graphite Introduced Polymer Membrane as a Shielding Layer in Ambient Temperature Sodium Sulfur Battery

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Herein, a strategy is demonstrated to effectively control the sodium polysulfide dissolution using composite polymer blend membrane. The sufficient interconnected pores in the white graphite introduced polymer blend membrane plays a multifunctional role as an ion selective membrane, evident from the ionic conductivity of 10^{-3} S cm^{-1} facilitating the movement of sodium ions and the presence of boron and nitrogen in white graphite acts as an effective absorbent to trap sodium polysulfides. The electrochemical testing reveals significant improvement of about 87.6% in specific capacity when compared to the cell without white graphite@ polymer membrane. Also, the cell exhibits an excellent cycling stability with a capacity retention of 83.1% at the end of 500 cycles. Other than the performance enhancement, the self-discharge battery is significantly reduced using the developed membrane. The better performance of the room temperature sodium sulfur battery is attributed to the physical and chemical confinement of composite polymer membrane which is also evident from the post analysis study. This is the first attempt of utilizing composite polymer membrane as a blocking layer in room temperature sodium sulfur batteries and is promising for the construction of high performance batteries.

1. Introduction

Lithium ion battery has emerged as the battery of choice for various portable electronic devices, electric vehicles, and grid storage applications.[1] The emergence of electric vehicles and grid storage application demands for the large-scale production of lithium ion battery. This poses a higher demand for lithium and the lithium reserves currently available is not sufficient to meet the needs.[2] In response to the increasing demand of rechargeable batteries, significant efforts have been devoted to develop a cost effective battery technology using widely available resources without compromising in specific capacity.[3,4] Looking for the post-lithium ion battery technologies, room temperature sodium sulfur batteries are considered to be a promising high energy density battery due to its high theoretical energy density and low cost due to the usage of inexpensive sodium and sulfur.[5–8] Though low cost and high energy density requirements are satisfied by sodium sulfur battery, there are several other drawbacks which limits the sodium sulfur battery to reach its level of commercialization. The foremost problem associated with sodium sulfur battery is linked to the generation and dissolution of polysulfides into the electrolytes. This dissolution of polysulfides not only causes the loss of active sulfur from the cathode, but also leads to pronounced shuttling effect and subsequently results in low energy efficiency of the battery. Moreover, this process accelerates the degradation of metallic sodium anode. Also, the poor electronic conductivity of sulfur (5×10^{-30} S cm^{-1}) and large volume expansion during cycling also decrease the utilization of active material and results in low life span of the battery.[9] Till now, few strategies have been reported to control the polysulfide shuttling in room temperature sodium sulfur battery. It has been demonstrated that polysulfide shuttling can be controlled by immobilizing sulfur inside the host material,[10–20] modification of separator,[21–26] and the use of additional layer between cathode and separator.[7,27,28] Immobilizing sulfur in the host material can control the polysulfide shuttling to a considerable extent due to the physical confinement of sulfur with the host material. Sulfur introduced in nitrogen-doped graphene as cathode in room temperature sodium sulfur battery is reported to show a specific capacity of 212 and 136 mAh g^{-1} at the end of first and tenth cycles, respectively. It is observed from this study that weak confinement of polysulfide with host cathode leads to very low specific capacity.[12] In addition, sulfur introduced in polyacrylonitrile nanofiber shows a specific capacity of 300 mAh g^{-1} at 0.01 C.[29] Since the dissolution of sodium polysulfides into the electrolyte is very much higher, the physical confinement using cathode host leads to failure of trapping sodium polysulfides. To overcome this problem, another approach of including polar metal oxides along with cathode support to

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utilize both physical and chemical confinement strategy to trap the sodium polysulfides is being reported[11,30,31] The use of interlayer is an alternative approach for constraining the polysulfides within the cathode structure. The use of carbon-based interlayer confines the polysulfides as well as acts a conductive path to the sulfur electrode and thus helps to obtain high rate capability and cyclic stability.[7] In addition, the optimization of electrolyte is also an alternative approach to control the polysulfide shuttling.[32] Though these techniques are proposed to limit the polysulfide shuttling, the fast decay in specific capacity caused by polysulfide shuttling is not fully addressed in room temperature sodium sulfur battery. The use of an ion selective membrane or solid electrolytes is considered to be an effective approach to prevent the dissolution of polysulfides compared to carbon-based interlayers.[33,34] The use of nafion, an ion selective membrane blocks the polysulfide anions and are therefore employed as a coating on the separator[33,35] as well as the solid electrolyte[28] between the electrodes. Though the use of nafion is demonstrated to control the polysulfide shuttling, the detailed study on control of sodium polysulfide shuttling is less focused.[36] The use of solid electrolyte like solid beta alumina membrane can control the sodium polysulfide dissolution but the ionic conductivity through the membrane at room temperature is a serious issue.[37] This urges us to explore about ion selective porous ceramic-based polymer membrane with the characteristics of both ionic conductivity and polysulfide trapping. In our previous work, $BaTiO_3$-based PVDF-HFP (poly (vinylidene fluoride-co-hexafluoro propylene))/(poly (butyl methacrylate)) PBMA blend is used as the separator for sodium ion battery and it exhibits ionic conductivity in the order of 10^{-3} S cm^{-1} at room temperature.[38]

So we have attempted to use PVDF-HFP/PBMA as the polymer matrix for ion selective membrane for the movement of sodium ions and the inclusion of inorganic materials into the polymer matrix increases the ionic conductivity.[39] Herein, we have chosen hexagonal white graphite (boron nitride, BN) as the inorganic filler which increases the ionic conductivity as well as immobilize the soluble polysulfides by the process of chemical confinement. So, white graphite incorporated PVDF-HFP/PBMA blend (white graphite @ polymer membrane)) is used as the shielding layer between cathode and separator. The white graphite @ polymer membrane has the dual benefits of blocking the polysulfides due to the presence of boron and nitrogen in white graphite and simultaneously allows only the sodium ions to migrate through the shielding layer. To utilize these benefits, room temperature sodium sulfur battery is designed using white graphite @ polymer membrane as the shielding layer between sulfur cathode and separator. The proof of concept is realized from the electrochemical study and the room temperature sodium sulfur battery delivers an improved specific capacity of 1135 mAh g^{-1} at 50 mA g^{-1} which is about 49% enhancement in specific capacity when compared to the cell without shielding layer. In addition, the cell is able to deliver a remarkable specific capacity of 500 mAh g^{-1} even at high current density of 2 A g^{-1}. Also, the cell is stable with the capacity retention of about 83.1% at the end of 500 cycles. It is also observed that the self-discharge rate of the battery due to the polysulfide dissolution is diminished using white graphite @ polymer membrane. This work demonstrates a new concept of chemical confinement of sodium polysulfides using polymer-based membrane in room temperature sodium sulfur battery and shows a great potential in the future application of room temperature sodium sulfur battery.

2. Results and Discussion

The surface morphology of polymer membrane and white graphite@ polymer membrane is shown in **Figure 1**a–d. The polymer membrane has a porous structure which provides better accessibility of electrolyte ions through the polymer membrane. The composite white graphite @polymer membrane also indicates the porous network with pore size of less than 10 µm as shown in Figure 1c,d. Compared to the pristine polymer membrane, the size of the pores on the surface of composite polymer membrane is reduced which also helps in preventing the dissolution of polysulfides into the electrolyte. Also, it is evident that white graphite in the composite polymer membrane is uniformly distributed over the polymer network. The uniform distribution of white graphite helps in strong chemical confinement of sodium polysulfides.

The energy dispersive spectroscopy (EDAX) mapping of white graphite@ polymer membrane indicates the uniform distribution of white graphite in the polymer matrix (Figure S1, Supporting Information). It is observed from the image that white graphite does not block the pores of the composite polymer membrane. This facilitates the movement of sodium ions through the composite polymer membrane. The corresponding cross-sectional image of white graphite @polymer membrane is shown in Figure S2 in the Supporting Information and the thickness of the composite membrane is observed to be 207.28 µm. As observed from the scanning electron microscopy (SEM) image, the pores of the membrane is not blocked by the addition of white graphite. This is further confirmed by the butanol uptake method. The porosity of the white graphite@ polymer membrane is found to be around 72% and for the polymer membrane it is about 58%. The addition of white graphite into the polymer membrane increases the porosity of the membrane which helps in the better electrolyte uptake which is consistent with the previous reports.[40–42]

The structural characteristics of the composite membrane is studied using X-ray diffraction (XRD) technique (Figure S3a, Supporting Information). The polymer membrane shows diffraction peaks at 18° and 20° indexed to (020) and (100) planes corresponds to α- and β-phases of PVDF-HFP, respectively. The diffraction peaks of white graphite shows peak at 26.6°, 41.6°, 43.7°, 55.1°, and 75.9° corresponding to (002), (100), (101), (004) and (110) planes respectively. The indexed peaks corresponds to the hexagonal crystal structure of BN. The appearance of characteristic BN peak in the composite polymer membrane indicates the presence of BN in the polymer matrix. Also, the surface morphology of white graphite is analyzed using SEM images as shown in Figure S4 in the Supporting Information. The surface morphology of white graphite shows sheet-like structure with the average particle size of about 103 nm. The good thermal stability of composite polymer membrane is necessary during the thermal run away of the battery and therefore analyzed using thermogravimetric analysis (Figure S3b, Supporting

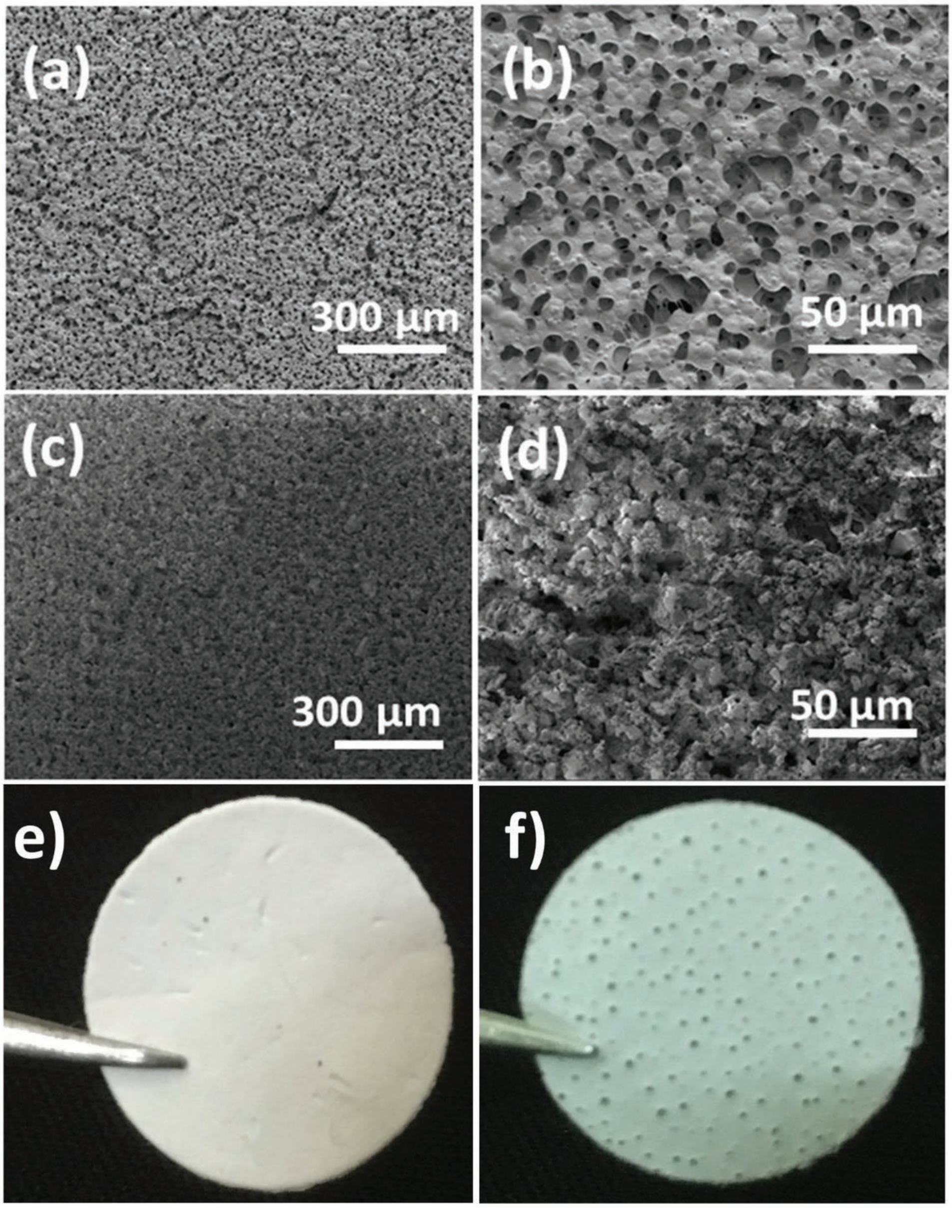

Figure 1. The SEM image of a,b) polymer membrane, c,d) white graphite@ polymer membrane, photographic image of e) white graphite@ polymer membrane, and f) polymer membrane.

Information). It is observed that weight loss occurs at a temperature from 216 °C to 430 °C due to the decomposition of polymers. Beyond that temperature, rapid weight loss occurs which is due to the complete degradation of polymers. After 500 °C, there is no weight loss is observed due to the presence of BN which is stable till 800 °C. In the composite polymer membrane, the quantity of BN is observed to be around 80 wt%. The presence of BN in the composite polymer membrane improves the thermal stability of the membrane as observed from the decomposition temperature. The good thermal stability of the membrane is sufficient for the safe operation during the thermal runaway of the battery. Also, the presence of high content of white graphite helps in controlling the polysulfide dissolution.

To demonstrate the trapping of sodium polysulfides using composite white graphite @polymer membrane, sodium sulfur battery is assembled using sulfur as the cathode and sodium metal as the anode in an ether-based electrolyte. The sulfur powder is coated on current collector with the loading of 0.6–0.8 mg cm^{-2} as shown in Figure S5a,b in the Supporting Information. In order to improve the conductivity of sulfur, partially exfoliated multiwalled carbon nanotubes are added as conducting additive. It is observed that the coating of sulfur is uniform and partially exfoliated multiwalled carbon nanotubes are uniformly distributed over the sulfur to provide a good conducting path (Figure S5b, Supporting Information). Also, the surface of the coating indicates the presence of pores which aids in better electrolyte penetration and facilitates the utilization of active sulfur. The conventional room temperature sodium sulfur battery uses separator soaked in liquid electrolyte between cathode and anode. With this design, the dissolved polysulfides can easily diffuse through the separator and reduces with sodium metal to form lower order polysulfides leading to the loss of active material. In this work, the composite white graphite @ polymer membrane is introduced as a shielding layer between the cathode and separator and the dissolution of active material into the electrolyte is prevented by the process of chemical confinement.

Other than the property of chemical confinement to be possessed by the shielding layer, the ionic conductivity of membrane at room temperature is a pre-requisite for an ion selective membrane to be used as polysulfide shielding layer. So the ionic conductivity measurements of white graphite@ polymer membrane and polymer membrane are carried out using electrochemical impedance spectroscopy technique. **Figure 2**a,b shows the ac impedance spectra and c, d) represents temperature dependent ionic conductivity of the membranes (with and without white graphite). In ac impedance spectra (Figure 2a,b), the appearance of inclined straight line and the absence of semicircle in the Nyquist plots implies that the charge carriers in the both the membranes are an ionic conductor (not an electronic conductor).[43] The impedance of the membrane is found to be decreasing with increase in temperature which is due to the expansion of polymer membrane at higher temperature. The bulk resistance value of white graphite incorporated polymer membrane varies from ≈13 to 8 Ω in the temperature range of 303–373 K and the impedance varies from ≈130 to 260 Ω for the polymer membrane in the temperature range of 303–373 K. The ionic conductivity of the membranes is calculated from the following Equation (1),

$$\sigma = \frac{L}{R_b\ A} \tag{1}$$

where, L is the thickness of membrane, R_b is the bulk resistance of the membrane (Figure 2a,b), and A is area of the membrane. Room temperature ionic conductivity of white graphite incorporated polymer membrane is found to be 1.134×10^{-3} S cm^{-1} and increases to 1.785×10^{-3} S cm^{-1} at 373 K. The ionic conductivity of polymer membrane is found to be 1.04×10^{-4} S cm^{-1} at room temperature and increases to 2.02×10^{-4} S cm^{-1} at 373 K. The ionic conductivity of both the membranes is observed to be increasing with increase in temperature which is due to the fact that membrane can expand easily and produces free volume which helps to improve the ionic mobility at higher temperature. The ionic conductivity of the polymer membrane (with and without white graphite) for the various temperatures are listed in **Table 1**. The white graphite incorporated in polymer membrane has higher ionic conductivity (10^{-3} S cm^{-1}) than the polymer membrane (10^{-4} S cm^{-1}). The incorporation of ceramic fillers in the polymer membrane helps for the structural rearrangement of the polymer and hinders the crystallization of the polymer. The presence of BN in polymer membrane increases the ionic conductivity and therefore can be used as an

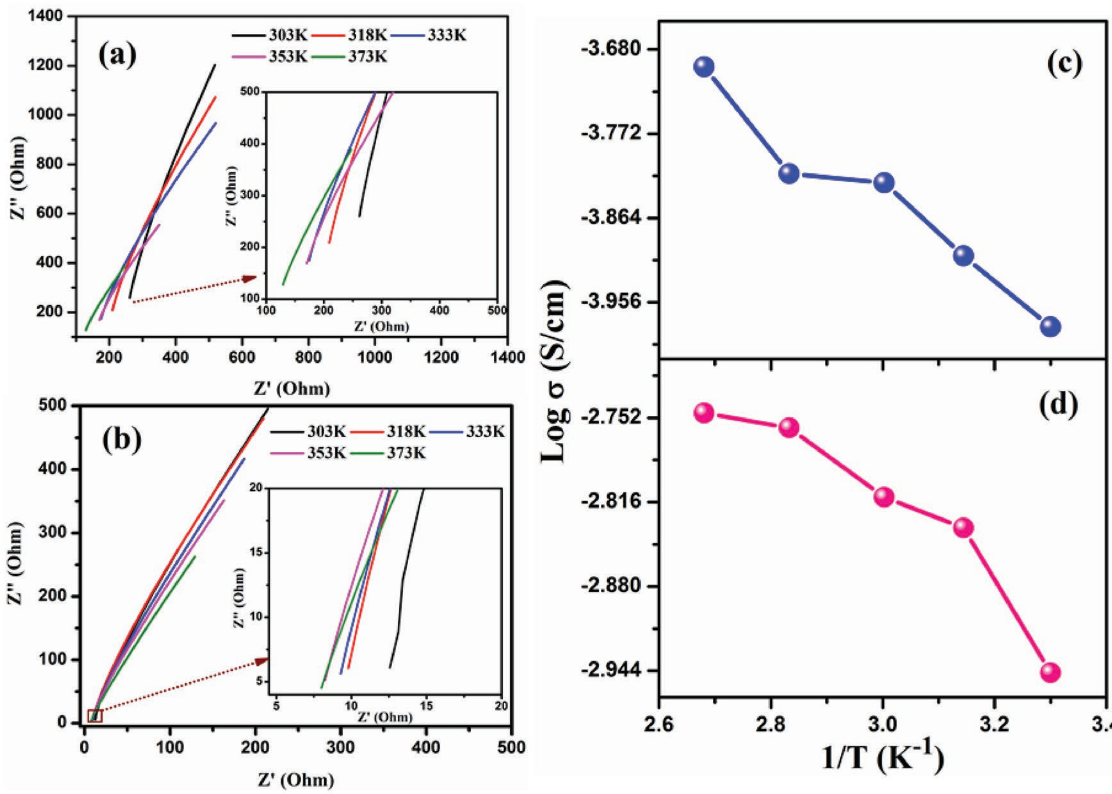

Figure 2. a,b) AC impedance plot of membranes (a. SS/polymer membrane/SS and b. SS/white graphite@ polymer membrane/SS) at different temperatures and c,d) temperature-dependent ionic conductivity of membranes (c. polymer membrane and d. white graphite@ polymer membrane).

Table 1. Ionic conductivity of the membranes at different temperature.

Sample code	Ionic conductivity × 10^{-3} [S cm^{-1}]				
	303 K	318 K	333 K	353 K	373 K
White graphite @polymer membrane	1.134	1.461	1.541	1.740	1.785
Polymer membrane	0.104	0.124	0.149	0.153	0.202

ion selective membrane. This permits the movement of sodium ions but simultaneously controls the diffusion of sodium polysulfides and thereby can be used as an effective shielding layer between the cathode and separator.

Due to the positive benefits, it will be interesting to evaluate the performance of sodium sulfur battery using white graphite@ polymer membrane. The electrochemical performance of room temperature sodium sulfur battery is studied using various electrochemical techniques. Initially, galvanostatic charge discharge measurements are carried out for the cells with and without shielding layer. The galvanostatic charge discharge curves of the cell with white graphite@ polymer membrane between the cathode and separator is shown in **Figure 3**a. The profile shows plateau as well as sloping region which is a typical charge discharge graphs of room temperature sodium sulfur battery. The presence of plateau as well as sloping region represents the reduction of elemental sulfur to short-chain polysulfide through the formation of long-chain sodium polysulfide. The charge discharge profile shows three plateaus during charging as well as discharging. During discharge, the plateau observed at 1.67 V corresponds to the reduction of sulfur S_8 to long-chain sodium polysulfide Na_2S_n (Na_2S_n $4 \leq n \leq 8$). The plateaus observed at 1.3 and 1 V corresponds to the reduction of long chain sodium polysulfide Na_2S_n to short-chain polysulfide Na_2S_m (Na_2S_m $2 \leq n < 4$) and then to Na_2S_2 respectively.[7] Further, Na_2S_2 is readily converted into Na_2S at 0.8 V. In the following process of charging, three plateaus are observed at 1.5, 2, and 2.2 V as shown in Figure 3a. The corresponding plateaus during charging represent the conversion of Na_2S to short-chain polysulfide, then to long-chain polysulfide and back to sulfur. The plateaus observed in galvanostatic charge discharge profile is stable even at higher current density of 2 A g^{-1}. This implies that fast reaction kinetics benefits from the presence of partially exfoliated multiwalled carbon nanotubes along with sulfur in the cathode. The cyclic voltammetry (CV) measurement is carried out in the potential range of 0.5–2.8 V at the scan rate of 0.1 mV s^{-1}. The CV curve exhibits three cathodic peaks centered at 1.7, 1.4, and 1.1 V which is consistent with the galvanostatic discharge voltage plateau regions. The corresponding anodic peaks are located at 1.6, 2, and 2.15 V. The presence of multiple oxidation and reduction suggests that sodiation and desodiation of sulfur occurs through multistep process. It is also evident that the peak intensity of sodium sulfur battery with white graphite@ polymer membrane is higher when compared to the cell without interlayer. This signifies the enhanced reaction kinetics due to the presence of white graphite and polymer membrane. To understand the role of polymer matrix, the polymer membrane without white graphite is used as the shielding layer and evaluated its electrochemical performance. It is clear from the CV curve that polymer membrane also helps in hindering the dissolution of sodium polysulfides. It is also been observed that there is a positive shift in the peak voltage during the anodic process and a negative peak shift during cathodic process as shown in Figure 3b and Figure S6 in the Supporting Information. The shift in the peak indicates the enhanced reaction kinetics for the conversion of sodium polysulfides. The sulfur cathode without the shielding layer delivers a specific capacity of about 551 mAh g^{-1} at a current density of 50 mA g^{-1}. As the current density is increased to 2 A g^{-1}, the specific capacity has reduced to 106 mAh g^{-1}. The low specific capacity at higher current density is due to the slow reaction kinetics of the polysulfide conversion. It is observed that the decay in capacity is incremented without interlayer and the capacity retention is observed to be 19.2% when the current is increased from 50 mA g^{-1} to 2 A g^{-1}. The polysulfide conversion efficiency as well as shielding effect of white graphite@ polymer membrane is evidenced from the superior rate capability of the cell (Figure 3c). The cell exhibits a reversible specific capacity of 1034 mAh g^{-1} at a current density of 50 mA g^{-1}. This value of specific capacity is about 53.2% enhancement when compared to the cell without interlayer at the same current density. The performance of the cell is also evaluated at a higher current density of 2 A g^{-1} and the corresponding specific capacity is about 389 mAh g^{-1}. The good electrochemical performance of the battery even at higher current density elucidates that reaction kinetics of polysulfide conversion reaction is enhanced with the inclusion of white graphite @ polymer membrane layer. This is consistent with the CV curve as the peak current increases in the cell with white graphite@ polymer membrane when compared to the cell without interlayer (Figure 3b). When the current is reduced back to 100 mA g^{-1}, a reversible specific capacity of 744 mAh g^{-1} can be recovered showing an excellent sodium storage and reversibility. Also, it is observed that the cell with polymer membrane exhibits better rate capability than the cell without interlayer. The presence of fluorine in the polymer membrane helps in the confinement of sodium polysulfides to a certain extent.[27] Though the presence of fluorine in the membrane can chemically confine the sodium polysulfides, the presence of larger pores in the membrane cannot control the complete migration of sodium polysulfides. With the addition of white graphite into the polymer membrane, the synergistic role of boron and nitrogen in white graphite and the controlled pore structure helps to achieve better electrochemical performance.

After the rate capability test, the cell is subjected for long-term cyclic stability at a current density of 500 mA g^{-1}. The cyclic stability curves show that the cell with white graphite @polymer membrane shows superior performance when compared to the cells with polymer membrane and without interlayer (Figure 3d). The cell with white graphite interlayer shows an initial specific capacity of 656 mAh g^{-1} after the rate

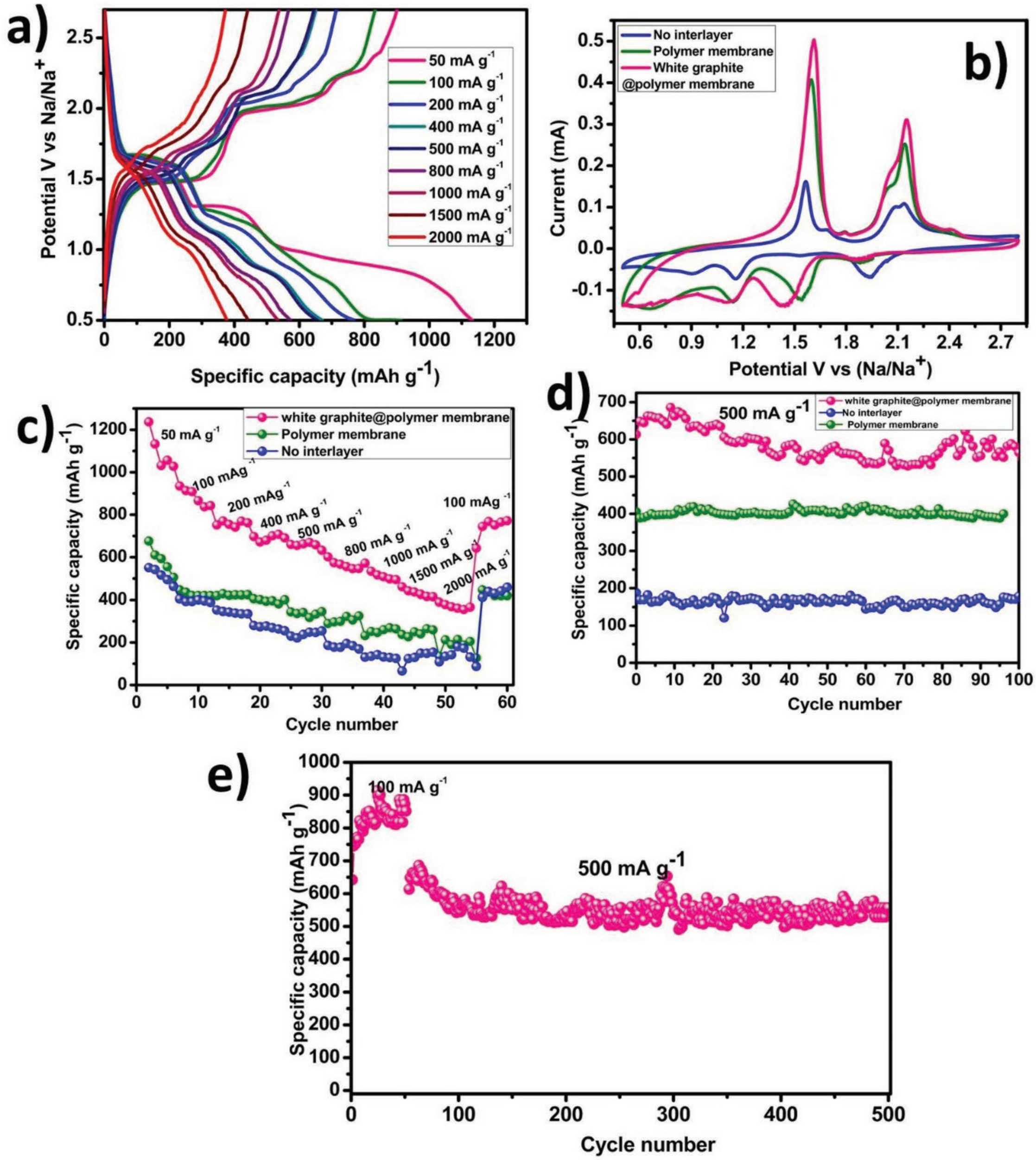

Figure 3. a) Galvanostatic charge discharge curves of white graphite@ polymer membrane, b–d) CV curves, rate capability and cyclic stability at 500 mA g^{-1} for sodium sulfur battery 100 cycles and e) cyclic stability of the cell with interlayer for 500 cycles.

capability test. At the end of 100 cycles, the specific capacity obtained is about 570 mAh g^{-1} with a capacity retention of about 88%. The enhanced cyclic stability is associated with an efficient control of sodium polysulfides by chemical bonding of boron and nitrogen present in white graphite with higher order sodium polysulfides. This is about 2.7 times enhancement in specific capacity when compared to the cell without interlayer. The long-term cyclic stability is carried out for the cell with white graphite @polymer membrane interlayer. In Figure 3e, the sodium sulfur cell has achieved a stable cycling of about 500 cycles with the first 50 cycles being cycled at 100 mA g^{-1}. The cell retains a specific capacity of 499 mAh g^{-1} at the end of 500 cycles. The good cyclic stability is attributed to synergistic role of the polymer membrane and the white graphite. The Coulombic efficiency of the sulfur-based batteries decides the extent of controlling the polysulfide shuttling. The Coulombic efficiency of the battery with white @polymer membrane, polymer membrane and without interlayer is shown in Figure S7 in the Supporting Information. It is observed from the figure that the cell with white graphite @polymer membrane exhibits 100% Coulombic efficiency which implies the control of polysulfide shuttling. But the cell without interlayer exhibits a fluctuating Coulombic efficiency with the minimum Coulombic efficiency of about 60%. Also, the cell with polymer membrane exhibits a Coulombic efficiency close to 100%. This further confirms the role of both polymer membrane and white graphite in controlling the polysulfide shuttling.

The self-discharge of metal sulfur battery is a serious issue when compared to lithium ion battery. So the self-discharge of the cell is tested by measuring the open circuit potential of the cell over the period of time (**Figure 4**a). In the first 3 d, the cell voltage tends to fall for the cell without interlayer which signifies the reduction of higher order polysulfides to lower order polysulfides.[28] Later, the voltage tends to increase due to the nucleation of Na_2S on the electrode surface. The open circuit potential of the cell with interlayer shows a steady potential of 2.35 V without drop in open circuit voltage. These results clearly signifies that white graphite@ polymer membrane assist in controlling the sodium polysulfide shuttling. The low self-discharge of the cell using white graphite @polymer membrane is further validated by using electrochemical impedance spectroscopy technique. The continuous dissolution of sodium polysulfide, increases the internal resistance of the cell. So, the electrochemical impedance spectroscopy technique is carried out as shown in Figure 4b,c in the frequency range of 10 mHz to 1 MHz. Before the measurement, the cell is discharged for few cycles and the impedance of the cell is carried out for the discharged cell each day. The Nyquist plots for the cells with interlayer show two semicircles with a depressed semicircle in the high-frequency region and a well-defined semicircle in the medium-frequency region with a sloping line in the low-frequency region. The depressed semicircle corresponds to charge transfer resistance related to the formation of solid electrolyte interface film formed between the electrodes and electrolytes, the well-defined semicircle corresponds to the resistance of the interface between electrode and interlayer.

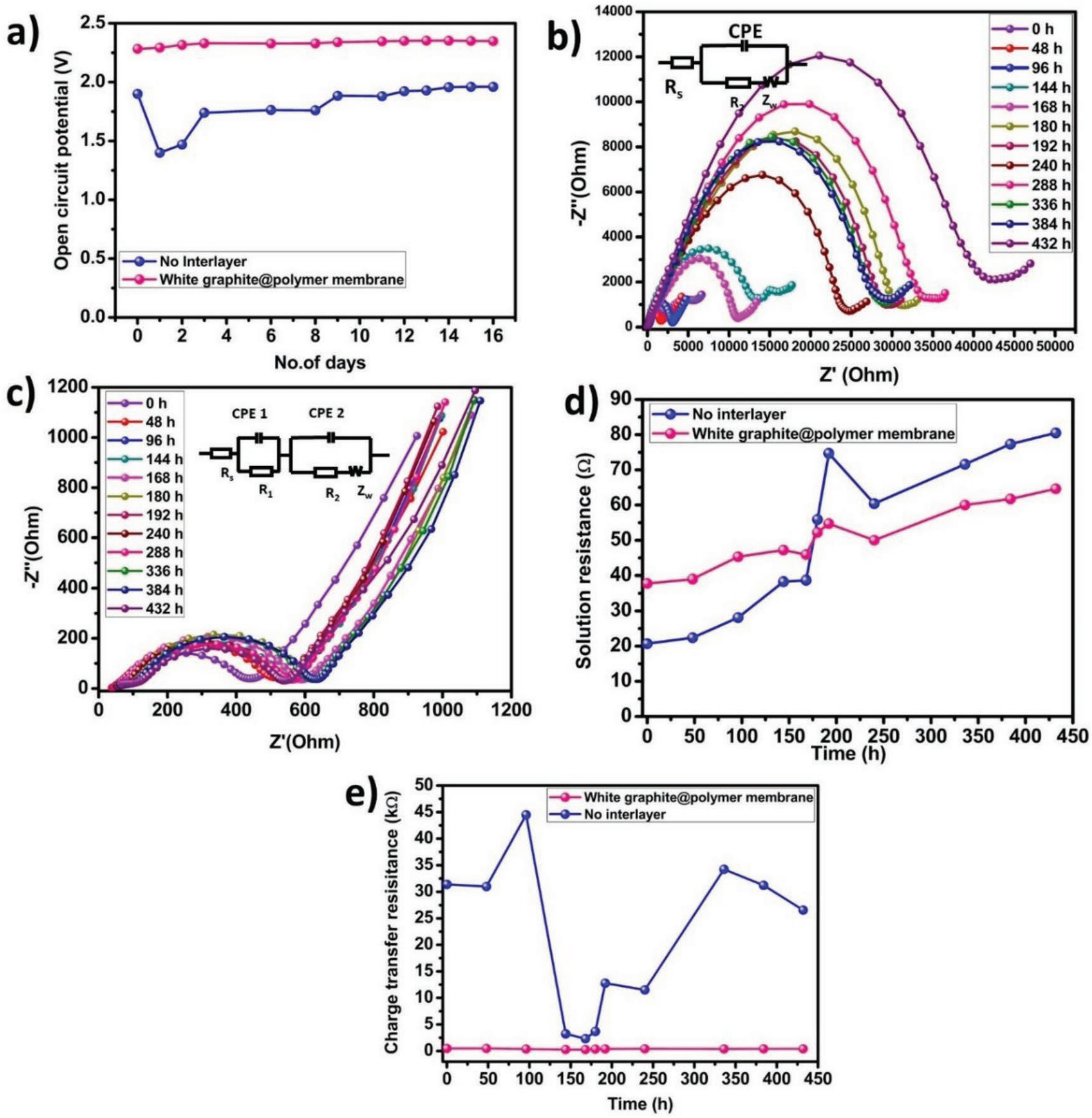

Figure 4. a) Change in open circuit potential of the cell as a function of time Long term impedance study of room temperature sodium sulfur battery with b) no interlayer, c) white graphite @polymer membrane, d) solution resistance, and e) charge transfer resistance.

The sloping line at 45° corresponds to the Warburg impedance arises due to the diffusion of sodium ions from the bulk of the electrolyte to the electrode surface. The cell without interlayer is fitted using the equivalent circuit with a semicircle and sloping region as observed from the Nyquist plot (Figure 4b). The impedance spectra is fitted using the equivalent circuit model as shown in Figure 4a,b, with R_s representing the solution resistance, R_{ct} corresponds to the charge transfer resistance, CPE 1 and CPE 2 are the equivalent model of double layer capacitance associated with the solid electrolyte interface film formed on the cathode and interlayer, respectively, and Z_w represents the diffusion controlled Warburg impedance related to the diffusion of sodium ions in the electrode.[44] Figure 4d represents the variation of solution resistance as a function of time and the solution resistance of the cell without interlayer is lower when compared to the cell with interlayer for a certain period of time. After 188 h, the solution resistance of the cell without interlayer, increases drastically to 80 Ω. Due to the lack of polysulfide confinement in the cell without interlayer, the dissolution of soluble polysulfides into the electrolyte increases leading to increase in the viscosity of the electrolyte. This causes increase in the solution resistance of the cell without interlayer.[45] Further, the confinement of polysulfide is also realized from the lower charge transfer resistance of the cell with white graphite @polymer membrane. The charge transfer resistance arises from the deposition of conversion products on the electrode surface. The charge transfer resistance remains to be constant of about ≈400 Ω over the period of time which signifies the role of white graphite interlayer in confining the soluble polysulfides (Figure 4e). The presence of cathodic interlayer effectively reduces the charge transfer resistance due to the binding of portion sodium polysulfides on the interlayer rather than getting deposited on the cathode structure. The very high value of charge transfer resistance for the cell without interlayer is due to the deposition of final discharged product on the electrode surface which acts as the barrier for the transport of ions. To confirm that, the impedance of the cell after assembly is carried out and the charge transfer resistance for the cell without interlayer is about 453 Ω (Figure S8, Supporting Information). So, the large value of the charge transfer resistance for the cell after charging implies that the process of polysulfide shuttling occurs in the cell. The deposition of polysulfide on the anode metal causes further reaction at the sodium anode leading to huge increase in the charge transfer resistance.[46]

In order to further understand the role of shielding layer in chemically binding the higher order sodium polysulfides, the surface morphology of sodium metal before and after cycling is characterized using SEM technique. It is assumed that the surface of sodium metal gets etched in the cell without interlayer due to the attack of polysulfides on the sodium metal. The digital photographic image of sodium metal shown in **Figure 5**a indicates that the surface of the fresh sodium metal looks smooth and glossy and the EDAX mapping shows only the presence of sodium and oxygen (Figure S9, Supporting Information). The SEM image of the fresh sodium metal appears to be plane surface with no deposits or cracks (Figure 5d). After cycling, the surface of the sodium metal contained in the cell with white graphite@ polymer membrane, is observed to be change in color compared to fresh sodium metal (Figure 5b) due to the interaction with the electrolyte. But from the SEM image it is observed that the surface is rough and are observed to be no deposits on the surface of the sodium metal (Figure 5e). In the case of cell without interlayer, the sodium metal surface is observed to be yellow in color due to the deposition and reduction of sodium polysulfides at the anode (Figure 5c). It is further analyzed from the SEM image that the surface of the sodium metal is observed to be corroded due to the attack of sodium polysulfides. Also, the encircled region in the image (Figure 5f) represents the presence of sodium sulfide deposits along with cracks and pits due to the attack of higher order sodium polysulfides. The corrosion of sodium metal anode is responsible for low specific capacity as observed from the charge discharge studies. To further confirm the deposits present on the sodium metal anode is due to polysulfide dissolution, the elemental mapping of the sodium metal for the cell with and without interlayer is carried out and are shown in Figures S10 and S11 in the Supporting Information. The elemental mapping of sodium metal contained in the cell without interlayer indicates the presence of sulfur on the surface of sodium metal which is a further confirmation of polysulfide dissolution. With the inclusion of white graphite @ polymer membrane, the diffusion of sodium polysulfide to the anode side is controlled as validated from the absence of deposits and corrosion on sodium metal. But in the elemental mapping of sodium metal from the cell with white graphite @polymer membrane (Figure S10, Supporting Information), traces of sulfur is observed which might be due to the dissolution in the initial few cycles as observed in the rate capability curve with the reduction in specific capacity in the initial two cycles. This demonstrates that sodium metal surface can be effectively protected by white graphite@ polymer membrane in the subsequent cycles. From this study, it is evident that shuttling effect is controlled due to the absorbability of the interlayer and are responsible for the enhanced specific capacity and cycle life even at higher current density. The absorbability of sodium polysulfides onto white graphite@ polymer membrane is examined by observing the morphology of the interlayer from the cycled cell. The surface of the fresh interlayer is observed to be the presence of uniform pores with the dispersion of white graphite over the polymer matrix (Figure 5g). After the process of cycling, it is evident that the surface of the interlayer is covered with agglomerates of particles which are predicted to be intermediate sodium polysulfide particles. From the high-resolution image as shown in Figure S12 in the Supporting Information, the polysulfides are covered over the surface of white graphite without blocking the pores of the interlayer and this implies the chemical bonding of polysulfides towards white graphite. The elemental mapping of the shielding layer represents the presence of both sodium and sulfur which confirms that sodium polysulfides are absorbed on the surface of interlayer (Figure 5i). Due to the shielding nature of white graphite @polymer membrane, enhanced specific capacity with good capacity retention is obtained.

In addition, the ability of white graphite @polymer membrane in suppressing the diffusion of sodium polysulfides is also visually evaluated (Figure S13, Supporting Information). The bottle filled with Na_2S_6 solution is placed in tetraethylene glycol dimethyl (TEGDME) solvent to compare the diffusion

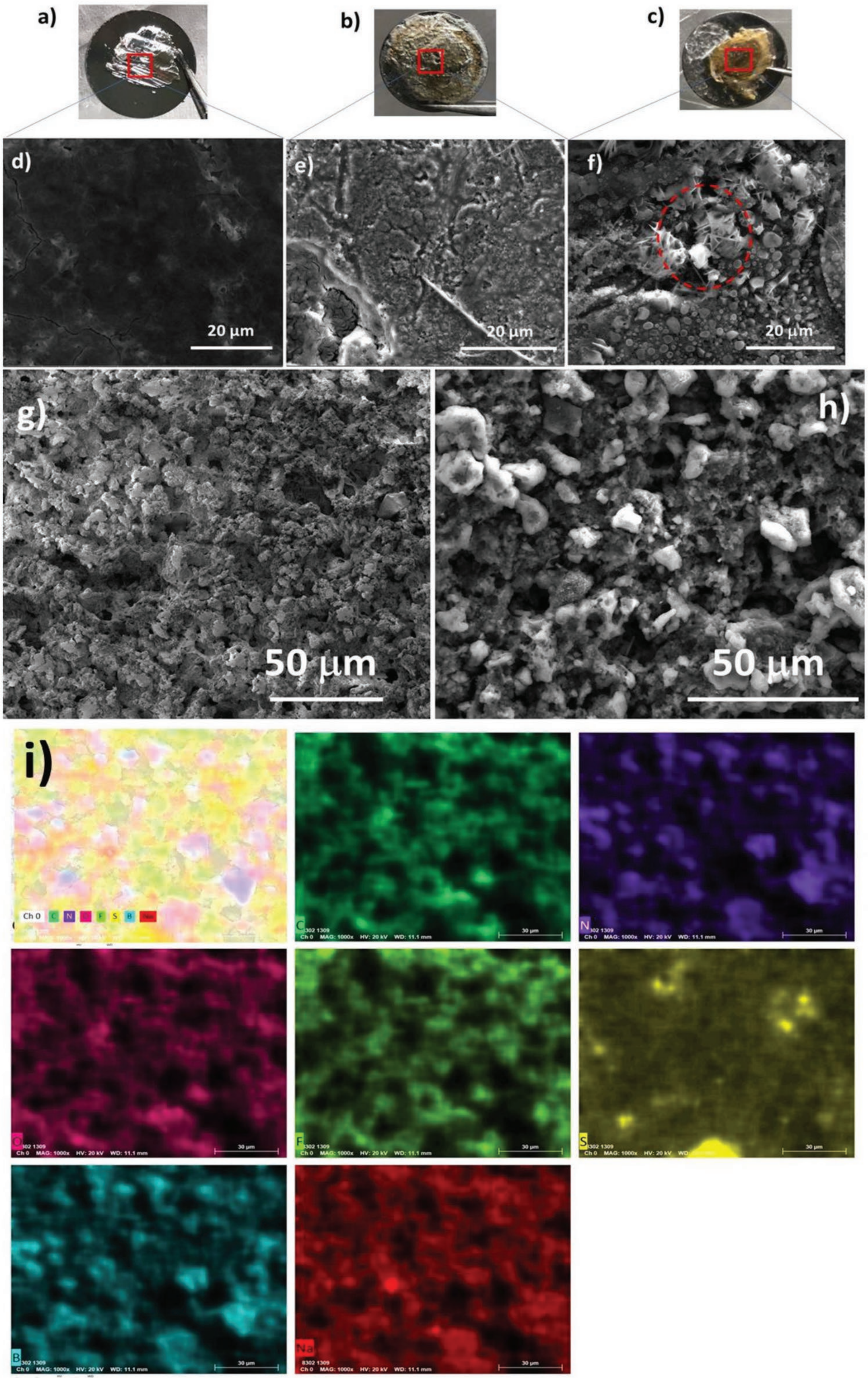

Figure 5. a–c) Photographic images, d–f) SEM images of fresh sodium metal, cycled sodium metal with white graphite @polymer membrane, no interlayer, SEM images of g) fresh interlayer, h) cycled interlayer, and i) EDAX mapping of cycled interlayer.

rate of sodium polysulfide through the separator. The glass bottle filled with Na_2S_6 solution has separator covering the top acting as a lid. Immediately after placing the bottle with sodium sulfides into the electrolyte solvent, the transparent nature of the solvent persist indicating no diffusion of sodium polysulfide. As the time increases to 12 h, the migration of polysulfides through the glass fiber membrane increases due to the concentration gradient as evident from the color change observed in TEGDME solvent. In contrast, the solvent in white graphite@ polymer separator, maintained the transparent nature indicating the controlled diffusion of polysulfides. As the time increases to 48 h, the electrolyte solvent has turned to dark brown in color which signifies the incapability of glass fiber membrane in inhibiting the diffusion of sodium polysulfides. At the same time, the electrolyte solvent remains to be transparent for the white graphite @polymer membrane even after 48 h. This clearly demonstrates that white graphite polymer membrane is effective in blocking the sodium polysulfides.

To further explore the efficiency of white graphite@ polymer membrane and nature of confinement of polysulfides towards the interlayer is further probed using X-ray spectroscopy technique (**Figure 6**). So, spectroscopy analysis of fresh and cycled interlayer in the discharged state is carried out to understand the interaction between sodium polysulfides and white graphite@ polymer membrane. The X-ray photoelectron spectroscopy (XPS) spectrum of fresh interlayer indicates the absence of sulfur and after cycling the presence of sulfur S 2p peak reveals the binding of polysulfide in the interlayer. The deconvoluted peaks of S 2p shows peaks at 162.2, 167.7,175.1, and 181.1 eV. The binding energy at 162.2 eV corresponds to the existence of polysulfide in the form of S_4^- indicating the existence of Na_2S_n on the surface of white graphite@ polymer membrane.[7] The binding energy at 167.7 eV corresponds to the interaction of electron negative sulfur species with nitrogen. The presence of peaks at higher binding energy corresponds to the presence of sulfate species.

The deconvoluted peaks of nitrogen N1s indicates the peaks at 401 and 403 eV for the fresh interlayer with the peaks shifted to 400 and 402 eV for the cycled interlayer. The peak at 401 eV corresponds to B—N bonding and the peak at 403 eV corresponds to the presence of Quaternary N peak. The nitrogen N1s spectrum of cycled interlayer is shifted to lower binding energy which corresponds to N—S bonding. The electronegativity of nitrogen helps to form covalent bond with sulfur. This further confirms that enhancement in performance is due to the chemical confinement of sodium polysulfides by white graphite present in the polymer membrane. The deconvoluted peak of B1s shows peak at 195.3 and 191.1 eV corresponding to N=B and N—B bonding respectively. Also, B 1s is observed to be shifted to higher binding energy in the case of cycled interlayer which implies that the role of boron also assist in binding sodium polysulfides. The existence of bonding of sulfur with boron and nitrogen is an evidence for preventing the migration of sodium polysulfides to the anode. Specifically, the presence of N=B and N—B in the composite polymer membrane causes the polarization of B and N atoms. This causes the charge attraction between B atoms and polysulfide anions thereby helps to trap the polysulfide much stronger.[47] In addition, the presence of both boron and nitrogen present in the composite polymer membrane acts as the anchoring sites for the polysulfides which thereby aids in enhancing the cyclic stability, specific capacity, and reduces the self-discharge of room temperature sodium sulfur battery.

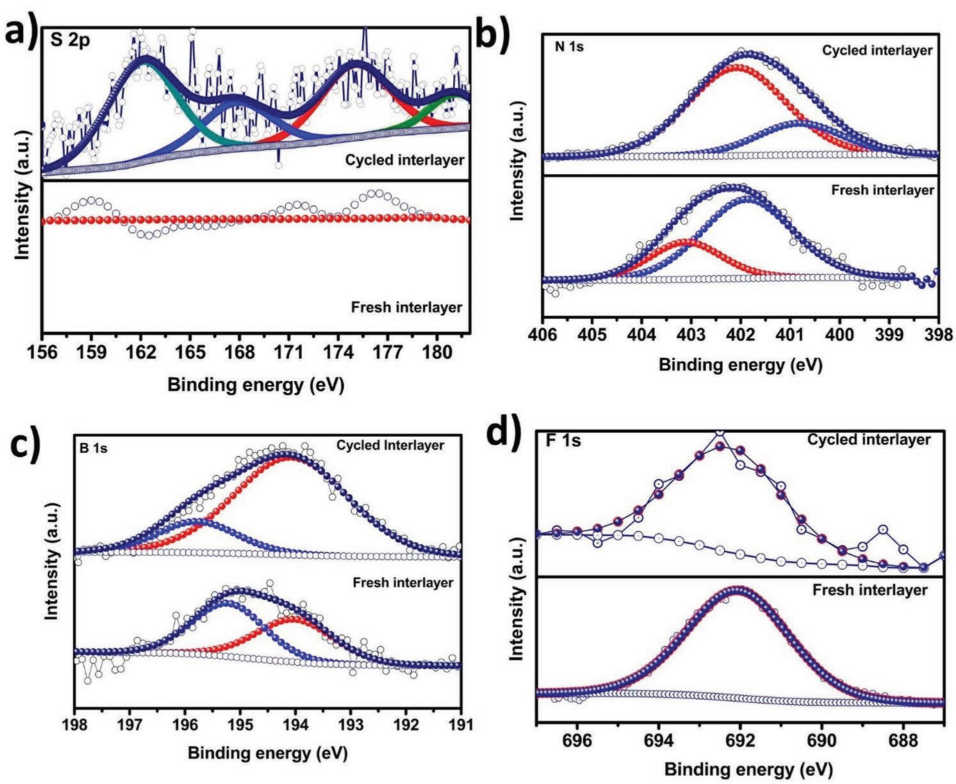

Figure 6. XPS analysis of fresh and cycled white graphite@ polymer membrane interlayer a) S 2p, b) N 1s, and c) B 1s and F 1s.

3. Conclusion

In summary, we have demonstrated an ambient temperature sodium sulfur battery with white graphite introduced blend polymer membrane as a polysulfide shielding layer between the cathode and separator. This membrane overcomes the polysulfide shuttling due to the chemical bonding of polysulfide with white graphite as well as acts an ion selective membrane for the movement of sodium ions. The operation of room temperature sodium sulfur battery with white graphite@ polymer membrane is demonstrated to deliver a specific capacity of 1034 mAh g^{-1} at a current density of 50 mA g^{-1} which is about 87.6% enhancement in specific capacity as well as stable capacity retention over 500 cycles when compared to the cell without shielding layer. The self-discharge behavior of the battery is significantly reduced by the use of white graphite@ polymer membrane and is evident from the impedance measurement wherein the polysulfide dissolution is controlled significantly. Also, the strong binding of polysulfide with boron and nitrogen present in the shielding layer is further confirmed from X-ray spectroscopy study. Thus, the developed membrane is far superior than the ion selective Nafion membrane in terms of cost as well as performance and can find potential application in room temperature sodium sulfur battery.

4. Experimental Section

Preparation of White Graphite @Polymer Membrane: PVdF-HFP {M_w = ≈400000 g mol^{-1}}, PBMA {M_w = 3 37000 g mol^{-1}} and white graphite (BN) were procured from Sigma–Aldrich. N,N-dimethylformamide and acetone were procured from Merck. Herein, polymer composite membrane was prepared by simple solution casting technique and the 70:30 ratios of white graphite (BN) and polymer was used. The required amount of substance (PVdF-HFP, PBMA, and BN) were dried at 100 °C by using vacuum oven (Thermofisher scientific) for 8 h to evaporate presence of moisture/impurities in the substance if any. Later, the dried polymers (PVdF-HFP: PBMA; 70:30 wt%) are dissolved in mixed solvent of dimethylformamide (DMF): acetone (3:7) and stirred for 10 h to get homogeneous solution. Further, the white graphite was incorporated into the homogeneous polymer solution and stirred for another 8 h to get uniform distribution of BN in the polymer matrix. Later, the obtained homogeneous slurry was casted on a glass plate and allowed for 24 h to evaporate solvents and finally get free standing polymer composite membrane. The obtained free standing membrane was dried in a vacuum oven at 60 °C for 10 h later which is transferred to the glove box for further characterizations. Furthermore, polymer membrane (without white graphite) was also prepared by same technique for the sake of comparison with white graphite incorporated polymer membrane.

Material Characterization: Powder XRD patterns were recorded using a Rigaku Smartlab X-ray diffractometer with Cu Kα radiation (λ = 1.5418 Å). All measurements were taken in the range of 10°–90° with the step size of 0.02. Thermogravimetric analysis was carried out in SDTQ600 analyzer from TA instruments in air and nitrogen atmosphere to 800 °C at a heating rate of 20 °C min^{-1} with a flow rate of 160 mL min^{-1}. XPS was carried out in Specs X-ray photoelectron spectrometer with Mg Kα as the X-ray source and PHOIBOS 100MCD analyzer. The SEM images were recorded using field-emission SEM Inspect F50 at acceleration voltage of 200 V to 30 kV. The samples were mounted via conductive carbon sticky tape. For the ex situ measurements, the interlayer is washed with TEGDME solvent and dried in an argon filled glove box and transferred immediately for XPS and SEM measurements.

Electrochemical Characterization: The electrode was fabricated by slurry coating process. The slurry was prepared by mixing 75% of sulfur, 10% of acetylene black, and with 15% of sodium carboxymethyl cellulose as binder in water solvent. The working electrodes were prepared by coating the slurry on the copper foil and then dried at 60 °C overnight in a vacuum oven. The dried electrodes were cut in the form of circular disc of 12 mm. 2032-coin cells were assembled in an argon-filled glove box where the oxygen and moisture level were maintained below 0.1 ppm using sodium metal as the counter electrode and glass fiber (Whatman GF/C) as the separator. The electrolyte used was 1 M $NaClO_4$ dissolved in TEGDME. The cells were galvanostatically cycled at various current densities between 0.5 and 2.8 V on Biologic BCS 810 battery cycler. CV were carried out on Biologic SP 300 in the potential range of 0.5–2.8 V at the scan rate of 0.1 mV s^{-1}. The areal sulfur loading is about 0.8–1 mg cm^{-2} and the volume of the electrolyte used is 15 µl.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

polymer membranes, room temperature, sodium sulfur, white graphite

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